

Design, Fabrication and Data Analysis of interferometers in Silicon Photonics

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Abstract: This report presents the modeling and data analysis of silicon photonics MZIs at 1550 nm. A corner analysis on a 500 x 220 nm waveguide boundaries the group index (ng) between 4.18 and 4.26, yielding an FSR of 5.64 to 5.75 nm for a Delta L of 100 μm . Layout parameters were verified across multiple configurations (Delta L up to 284.02 μm). Experimental characterization evaluates the transmission spectral response, showing a distinct correlation between the designed arm length differences and the resulting interference fringe density, as well as directional routing behaviors.

1. Introduction & Theory

This report focuses on the design, modeling, and experimental characterization of integrated Mach-Zehnder Interferometers (MZIs) in Silicon Photonics. The primary focus is placed on transverse electric (TE) mode operation at a central wavelength of 1550 nm.

2. Modeling and Simulation

A manufacturing corner analysis was performed on a nominal 500 x 220 nm strip waveguide for TE polarization at 1550 nm to evaluate the impact of process variations (wafer thickness: 215.3 - 223.1 nm; waveguide width: 470 - 510 nm).

Group Index (ng) Range: * Minimum ng: 4.18

Maximum ng: 4.26

Free Spectral Range (FSR) Variation: For an MZI configuration with an arm length difference of Delta L = 100 μm :

Minimum FSR: 5.64 nm

Maximum FSR: 5.75 nm

Based on that waveguide a variety of photonic circuits was developed and simulated by Ansys Lumerical. The standard setup for the MZIs can be seen in Fig. 1.

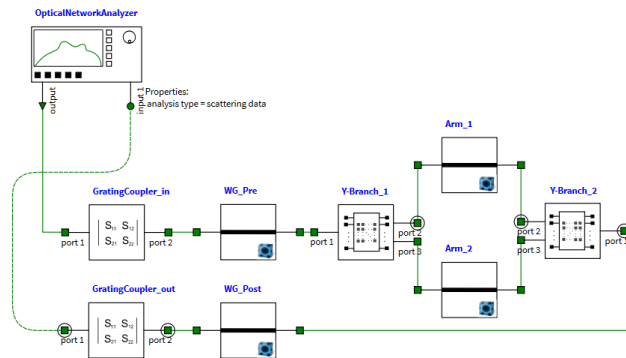


Fig. 1. Photonic Circuit design of MZI in Ansys Lumerical.

3. Design Parameters and Layout

The experimental chip layout includes multiple MZI configurations with layout parameters defined in Table 1. The arm length difference ($\Delta L = |\text{Arm}_1 - \text{Arm}_2|$) determines the device spectral response.

Table 1: MZI Layout and Design Parameters

Name	WG_Pre length	WG_Post length	Arm_1 length	Arm_2 length	Arm difference
MZI21/MZI22	159.81	102.44	111.65	395.67	284.02
MZI12	40.91	183.44	115.68	303.86	188.18
MZI11	40.91	136.07	115.68	303.86	188.18
MZI02	0	0	138.00	242.00	104.00
MZI01	0	0	245.10	390.71	145.61

Fig. 2 shows all fabricated circuits. In this work we focus on MZI 2.1 and MZI 2.2 in Fig. 3 and three additional MZIs 11, 01 and 02 (Fig. 4).

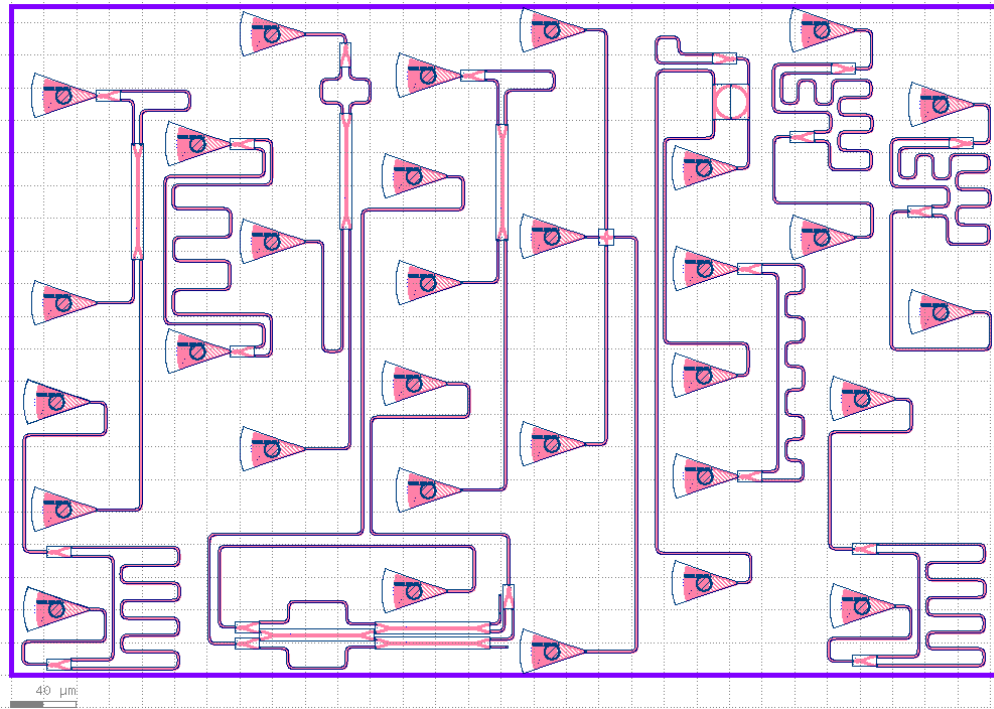


Fig. 2. Chip Layout.

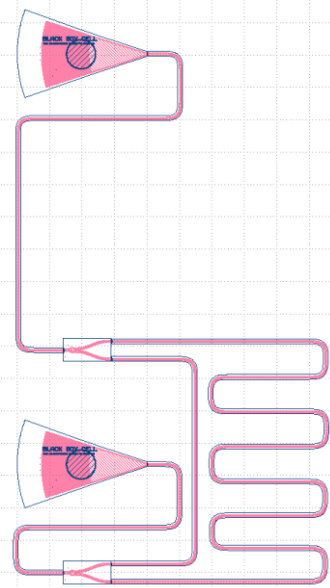


Fig. 3. Chip layout of MZI 21/22.

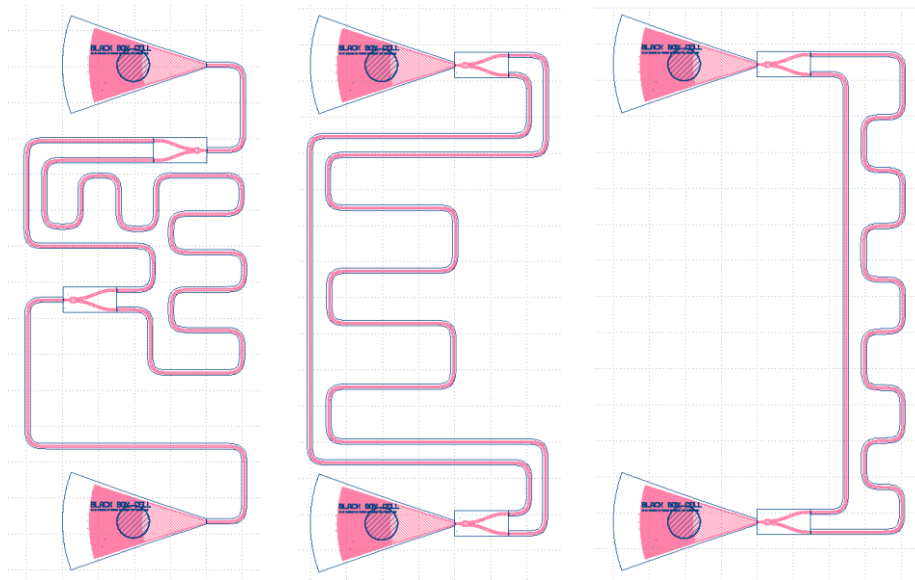


Fig. 4. Chip layout of MZI 11, MZI 01 and MZI 02

4. Experimental Data & Setup Configuration

The experimental transmission spectra were recorded using a multi-detector setup (Detectors 1, 2, 3, and 4) sweeping across a wavelength range from 1500 nm to 1600 nm. The optical power is measured in dBm. The output characteristics show the classic envelope shape governed by the grating coupler transmission profile, peaking near 1545-1550 nm, combined with high-frequency Fabry-Perot oscillations or noise floor limits at the spectrum edges (below -60 dBm).

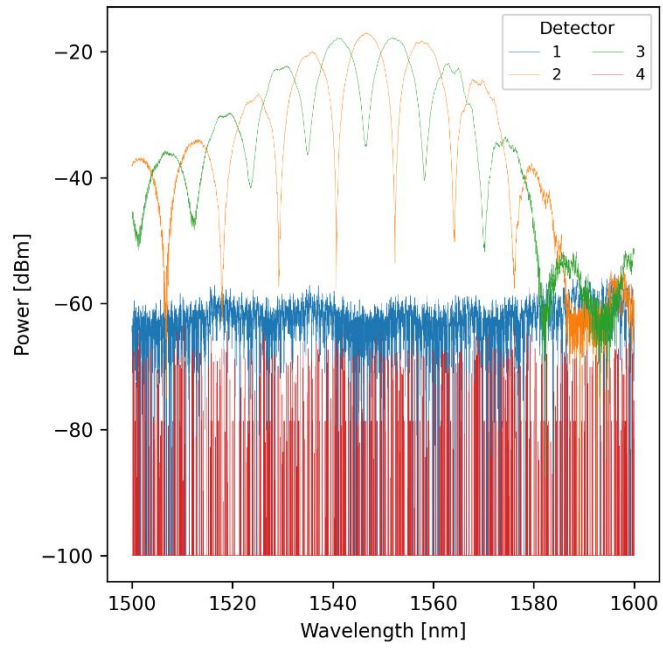


Fig. 5. Measured transmission spectrum of MZI21 (Delta L = 284.02 μm).

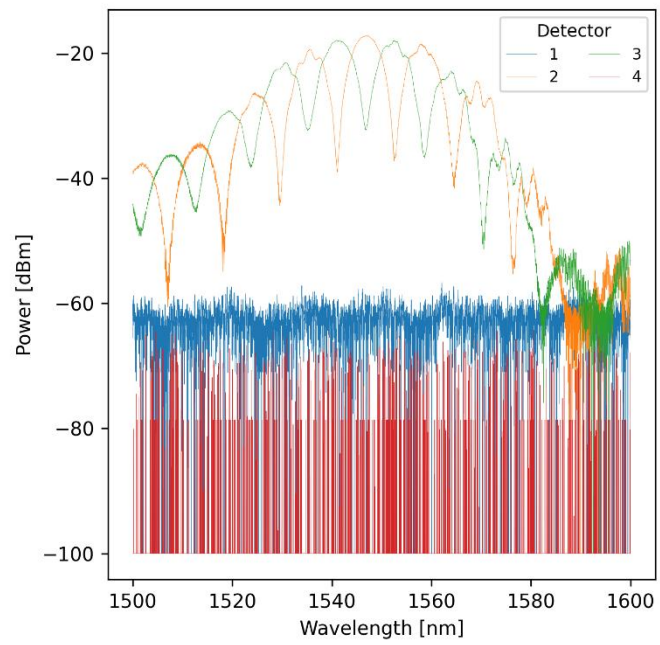


Fig. 6. Measured transmission spectrum of MZI22 (Delta L = 284.02 μm).

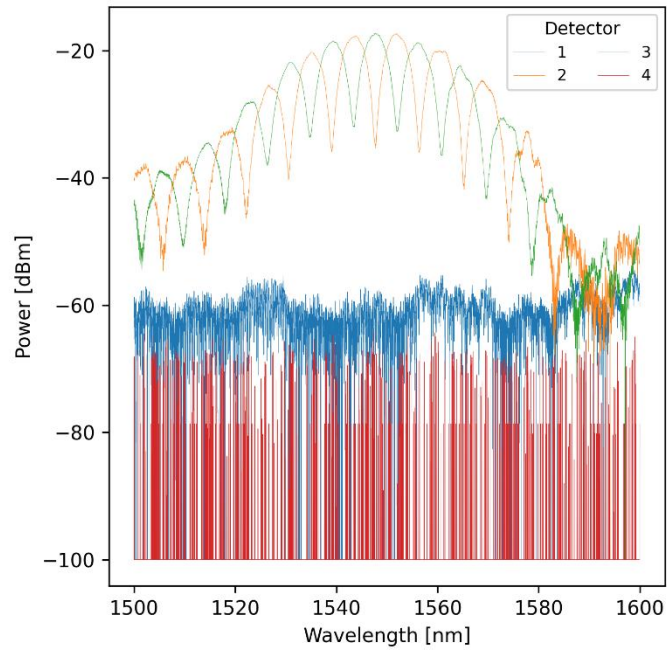


Fig. 7. Measured transmission spectrum of MZI11 in forward configuration.

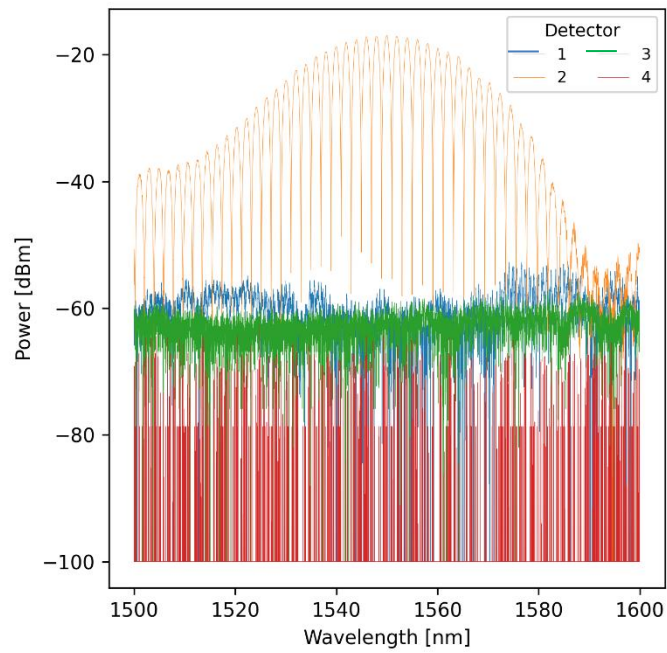


Fig. 8. Measured transmission spectrum of MZI11 in backwards configuration.

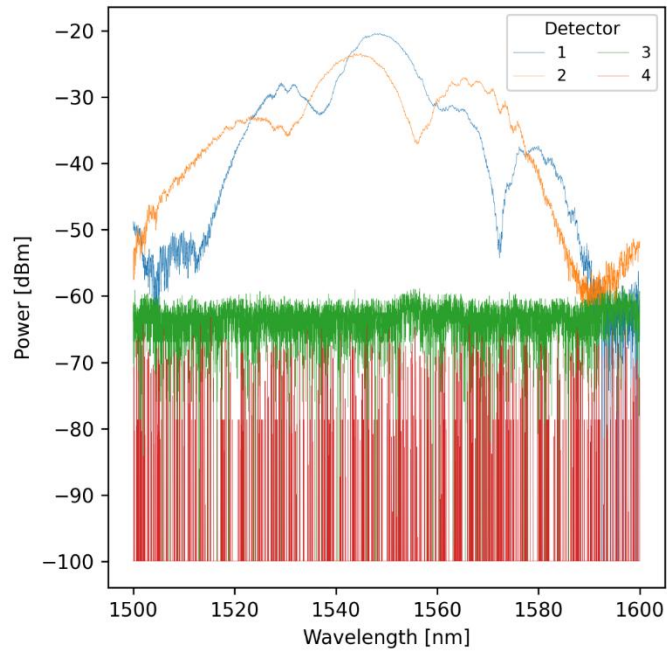


Fig. 9. Measured transmission spectrum of MZI01 (Delta L = 145.61 μm).

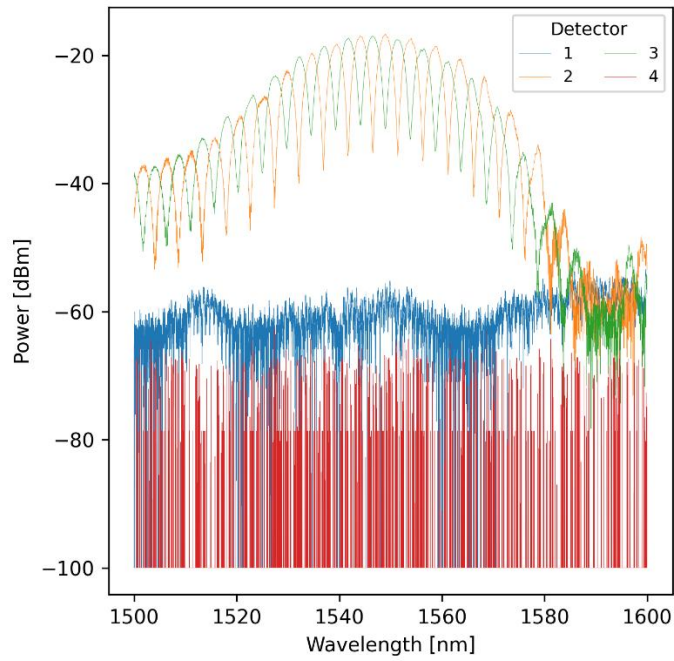


Fig. 10. Measured transmission spectrum of MZI02 (Delta L = 104.00 μm).

5. Data Analysis and Dataset Comparison

The measured transmission curves show a precise correlation with the underlying layout geometries from Table 1:

- **MZI21 and MZI22 (Delta L = 284.02 μm):** Due to having the largest arm length difference, these structures exhibit the highest fringe density (smallest Free Spectral Range) in the transmission spectrum. Detectors 2 (orange) and 3 (green) display perfectly phase-shifted, complementary interference patterns with peak power levels reaching approximately -18 dBm near 1545 nm. Detectors 1 and 4 capture noise at the -60 dBm to -100 dBm floor, indicating successful routing to the active ports.
- **MZI11 Forward vs. Backwards (Delta L = 188.18 μm):**
 - In the **Forward** direction, Detectors 2 and 3 receive the complementary modulated signals with an intermediate fringe spacing that directly matches the smaller Delta L compared to MZI21/22.
 - In the **Backwards** direction, the routing changes significantly. The interference pattern is exclusively captured by Detector 2 (orange) with high transmission, while Detector 3 (green) drops completely into the noise floor at -60 dBm. This confirms directional port isolation or asymmetric circuit behavior when fed from the opposite side.
- **MZI01 (Delta L = 145.61 μm):** The spectrum reveals a broader interference period (larger FSR) due to the reduced arm mismatch. Detectors 1 (blue) and 2 (orange) track the main signal components, showing a clean sinusoidal modulation overlapping the grating coupler envelope.
- **MZI02 (Delta L = 104.00 μm):** Having the smallest non-zero arm difference among the asymmetric designs, MZI02 exhibits the widest fringe spacing (largest FSR) across the 1500-1600 nm band. Detectors 2 and 3 capture the distinct wide oscillations with high peak power values around -17 dBm.

6. Conclusion

The analyzed silicon photonics devices successfully demonstrate Mach-Zehnder interferometric operation. The experimental Free Spectral Range scales inversely with the calculated arm differences (Delta L) from Table 1, ranging from the highly dense oscillations of MZI21/22 down to the wide fringes of MZI02. Furthermore, the comparison between MZI11 forward and backward configurations verifies robust directional routing behavior within the active multi-detector network.